

A Support–Faithfulness Contrast Aligns Across Model Families up to a Linear Map but Does Not Transfer to Real Hallucinations: A Matched-Pair Study

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Abstract

Internals-based, reference-free hallucination detection treats “context-faithfulness”—is an answer entailed by the provided context?—as a linear direction in the residual stream. But in the standard extractive setup, faithfulness is almost perfectly correlated with *lexical overlap* (whether the answer string occurs in the context, $r=0.99$ in our data), so an apparent “grounding probe” may be a string-matching detector. We introduce a matched SUPPORT×OVERLAP construction that decouples the two ($r=0.50$ on its model-free cells), including a wrong-question trap on which an off-the-shelf NLI model is fooled $\sim 25\%$ of the time. With this control, on four instruction-tuned families (Llama-3.1-8B, Qwen2.5-7B, Mistral-7B, Gemma-2-9B), we report: (i) a mid-layer context-grounding signal *does* exist beyond lexical overlap—within the answer-present stratum, where overlap is held constant, support is decodable at AUROC 0.87 (raw) and 0.72–0.78 after purging model confidence—though by construction this stratum cannot separate grounding from factoid-correctness, so we make no claim of a grounding axis geometrically distinct from factuality. (ii) This direction *aligns across the four families up to a linear map*: an orthogonal-Procrustes map transports it nearly collinear with each family’s native direction (cosine ≈ 0.98 , vs. ≈ 0.01 for a random direction through the same map), and far exceeds an anchor-projection baseline; we rest this on the geometry rather than the (ceiling-saturated) transferred-probe AUROC. (iii) Yet the synthetic direction *does not* transfer to real hallucination detection: on RAGTruth it scores below NLI, confidence, and lexical overlap among single-pass API-free baselines, and within a task no internals- or NLI-based signal (including a sentence-level SummaC-style NLI baseline) exceeds ≈ 0.6 AUROC—while overlap’s apparent edge (0.753 pooled, 0.647 macro) is largely a between-task base-rate artifact. Only an in-domain probe trained on hu-

man labels recovers a usable signal (0.77–0.82 on the three RAGTruth families; it exceeds a single-hypothesis NLI baseline but only ties lexical overlap on one). Our contribution is a confound-controlled methodology, a released wrong-question diagnostic, and a cautionary result: a grounding direction identified on controlled stimuli is real and portable, but is not, by itself, a deployable hallucination detector. All experiments are API-free and run on a single A100-40G.

1 Introduction

A grounded language model should answer from the *provided context*, not from its parametric memory or from surface cues. Reference-free, internals-based methods operationalize this by reading a linear “faithfulness” or “truth” direction from the residual stream (Azaria and Mitchell, 2023; Marks and Tegmark, 2024; Bürger et al., 2024; O’Neill et al., 2025), an attractive recipe because it needs no external judge and costs roughly one forward pass. Two distinct notions are routinely conflated under this banner: *parametric-factuality*—is a claim true in the model’s memory?—and *context-faithfulness*—is a claim entailed by the given context? Recent work treats both as activation objects and asks how they relate (Wang et al., 2025), but samples the two notions from *different* datasets, so any measured geometry is confounded with topic and lexicon.

We identify a sharper confound that undermines the faithfulness side specifically. In the standard extractive QA setup used to build such probes (e.g. entity-swap contexts (Longpre et al., 2021)), an answer is context-faithful *iff* the answer string occurs in the context. Faithfulness is therefore almost perfectly correlated with *lexical overlap* ($r=0.99$ in our data), and a probe that appears to read “grounding” may simply detect string copying. Indeed, after regressing out lexical overlap, an apparently excellent grounding probe collapses to chance.

A matched control. We introduce a SUPPORT×OVERLAP construction (§3) that crosses semantic support with lexical presence in a balanced 2×2 . Two cells are model-free: a *wrong-question trap*, where the answer is present but answers a different question (supported=no, overlap=yes), and the usual unsupported-absent cell; a paraphrase cell closes the design. This drops $\text{corr}(\text{SUPPORT}, \text{OVERLAP})$ from 0.99 to 0.50, and—tellingly—an off-the-shelf NLI model is fooled by the clean wrong-question trap $\sim 25\%$ of the time, confirming that lexical overlap misleads even “gold” entailment checks.

Contributions. With this control we run a careful study across four model families (Llama-3.1-8B, Qwen2.5-7B-Instruct, Mistral-7B-v0.3, Gemma-2-9B). We report two positive results and two honest negatives.

1. **An overlap-independent grounding signal (§5.1).** A mid-layer grounding signal *does* exist beyond lexical overlap: it decodes support among answer-present items at AUROC ≈ 0.87 (raw; 0.72–0.78 after purging model confidence) on all four families under item-grouped cross-validation. We are explicit about a limit of the design: within the answer-present stratum, support and our *synthetic* factuality label are identical by construction, so this isolates grounding from *lexical overlap* but *not* from factoid-correctness, and we do not claim a grounding axis geometrically distinct from a parametric-factuality direction.
2. **Cross-family alignment up to a linear map (§5.3).** An orthogonal-Procrustes map carries the grounding direction between families so the transported direction is nearly collinear with the target family’s *native* direction (cosine ≈ 0.98 , vs. ≈ 0.01 for a random direction through the same map), whereas an anchor-projection transfer reaches only 0.67 AUROC (Kim and Han, 2026). The transferred-probe AUROC (0.91) equals the within-family bound but is ceiling-saturated, and signed map-aware nulls sit at chance (0.51); we therefore rest the claim on the cosine. The factuality axis transports identically, indicating a shared low-rank contrast rather than a grounding-specific axis; we claim no universality beyond these four 7–9B models (Puri et al., 2025).
3. **No confidence asymmetry (§5.2).** Contrary to

a natural hypothesis, softmax confidence and activation norm do not absorb the factuality axis more than the faithfulness axis; the effect is null.

4. **Synthetic axes do not transfer to real hallucinations (§5.4).** Applied to RAGTruth (Niu et al., 2024), the controlled-stimulus grounding direction scores *below* NLI, confidence, and lexical overlap among single-pass API-free baselines; within a task no internals- or NLI-based signal—including a sentence-level SummaC-style NLI baseline—exceeds ≈ 0.6 AUROC, and even lexical overlap, the strongest baseline, reaches only macro 0.647. An in-domain probe trained on RAGTruth human labels recovers a usable signal (0.77–0.82 on the three RAGTruth families; exceeds a conservative single-hypothesis NLI baseline but ties lexical overlap on Qwen), so a real grounding signal exists in the internals but is a *different direction* from the synthetic axis and needs in-domain annotation.

The takeaway for the “faithful and efficient” agenda is cautionary but useful: internals-based grounding directions identified on controlled data are real and remarkably portable across model families, yet are *not*, by themselves, deployable hallucination detectors—on real data, lexical overlap still dominates.

2 Related Work

Geometry of truth and faithfulness. Linear “truth” directions are decodable from activations (Azaria and Mitchell, 2023; Marks and Tegmark, 2024) and partly shared across models (Bürger et al., 2024), but are task-dependent and often fail to transfer across tasks (Azizian et al., 2025; Bao et al., 2025). Wang et al. (2025) treat factuality and faithfulness hallucinations as activation subspaces and argue they overlap; crucially they draw the two notions from different corpora, leaving the geometry confounded with surface form. Adarsh et al. (2026) study how a single truth vector rotates when context is added. We instead hold the two notions on *matched* stimuli and control lexical overlap explicitly; we note, however, that our factuality label is a synthetic gold-answer relabeling rather than a parametric memory-truth probe, so we engage this line methodologically without claiming to adjudicate whether factuality and faithfulness share a direction.

Knowledge conflict and source attribution.

Probes can detect knowledge conflict and predict which source a model follows (Zhao et al., 2024; Sun et al., 2025; Brink et al., 2026). These are behavioral/source signals; we ask whether a *content* grounding axis exists that is not reducible to lexical overlap.

Cross-model transfer. Linear concept directions can be aligned across models via orthogonal Procrustes (Puri et al., 2025) or anchor projection (Kim and Han, 2026). We apply both to a grounding direction; Procrustes renders it nearly collinear with the target’s native direction (cosine ≈ 0.98) whereas anchor projection does not.

Reference-free hallucination detection.

NLI/entailment and alignment scores (Zha et al., 2023), single residual directions (O’Neill et al., 2025), confidence/correctness probes (Cho et al., 2026; Moreno Cencerrado et al., 2025), and faithfulness-aware UQ (Fadeeva et al., 2025) all target RAG faithfulness. We use NLI, confidence, and lexical overlap as baselines on RAGTruth and find our controlled-stimulus axis does not beat them.

3 The SUPPORT $\times$ OVERLAP Construction

We build matched statements from NQ-Swap (Longpre et al., 2021), which provides, per question q , a gold answer a_{true} supported by an original context c_{org} and a substituted entity a_{swap} supported by an entity-swapped context c_{sub} . We define two bits per (context, answer): SUPPORT S (does the given context entail the asserted answer for the asked question?) and OVERLAP O (does the answer string occur in the context?). The factorial uses four cells:

cell	(question, context, answer)	S	O
A	$(q, c_{\text{org}}, a_{\text{true}})$	1	1
B	$(q, \text{paraphrase}(c_{\text{org}}), a_{\text{true}})$	1	0
C	$(q', c_{\text{org}}, a_{\text{true}})$	0	1
D	$(q, c_{\text{sub}}, a_{\text{true}})$	0	0

Cell C is a *wrong-question trap*: a_{true} is present in c_{org} but is offered as the answer to a different question q' (the next item’s question). Cell B paraphrases c_{org} so the answer is entailed but not copied (open instruct model, kept only if token-overlap ≤ 0.34 and an NLI model still entails it); in this release cell B has *zero* yield, so all reported analyses use the model-free cells A/C/D with $\text{corr}(S, O) = 0.50$ (0.57 after removing the donor-collision items below; an orthogonalized

$\text{corr} \approx 0$ design would require cell B). One caveat in cell C: NQ-Swap dev has only 418 distinct questions over 1,200 items, so for 278 items (23.2%) the donor q' is string-identical to q and a_{true} *does* answer it; we flag these *donor-collision* items and report cell-C results with them removed (the full construction will dedup $q' \neq q$). On *clean* cell C an open NLI model (He et al., 2023) still rates 25.5% of unsupported-but-present items as “supported” (mean entailment 0.31, vs. 0.74 for cell A and 0.18 for cell D); the raw 37.6% rate is inflated by the collisions, which NLI entails 77.7% of the time (correctly). Lexical overlap thus misleads “gold” entailment even after the label error is purged (Fig. 1).

4 Probing, Separability, and Transfer

Directions. For a (model, layer) we extract last-token residual states X over the asserted answer span (one forward pass, bf16, left-padded). A SUPPORT direction d_S and FACTUALITY direction d_F are mass-mean (difference-of-means) estimates (Marks and Tegmark, 2024); signed projections Xd are the probe scores. We report layer-wise cross-validated AUROC under *GroupKFold* on ITEM_ID: because cells A/C/D of an item reuse the same context, an ungrouped split would leak near-duplicate rows across folds, so all statements of one item are kept in a single fold. Axis polarity is fixed on the *train* fold; the test projection is scored as-is (no $\max(\text{AUC}, 1 - \text{AUC})$ flip), so a non-predictive direction correctly averages to 0.5 rather than being floored above it. All point estimates carry item cluster-bootstrap 95% CIs.

Overlap control. To test grounding *beyond* overlap we (i) restrict to the answer-present stratum $O=1$ (cells A vs. C), where lexical presence is constant, and (ii) residualize X on surface covariates (overlap, length) before re-probing.

Null-calibrated separability. We treat each notion as an undirected axis and measure the acute angle $\angle(d_S, d_F)$. Significance is assessed against a within-notion null built by splitting one notion’s data in half and measuring the half-vs-half angle; a permutation test with $n_{\text{null}}=10,000$ draws gives p . We report the resolution bound $p < 1/n_{\text{null}} = 10^{-4}$ rather than $p=0$ when no null draw exceeds the observed angle, and state that the test is run per family at the a-priori mid layer (four families $\times$ one layer).

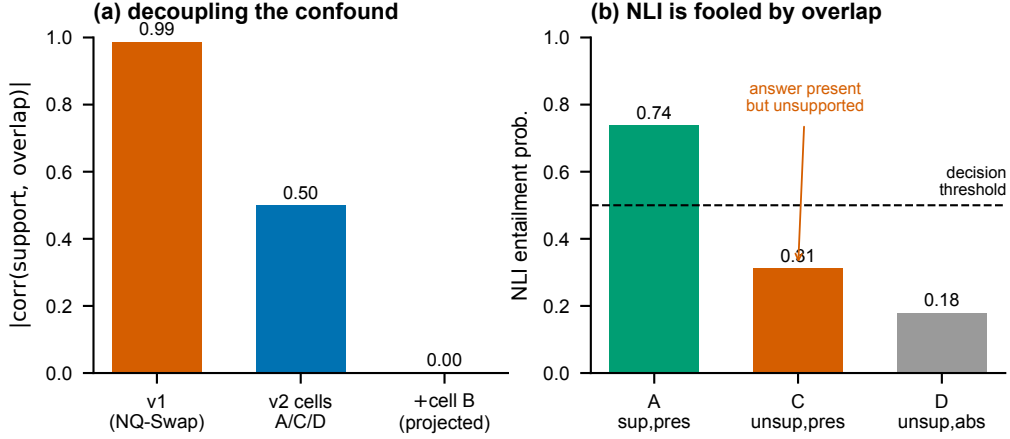


Figure 1: **Decoupling grounding from lexical overlap.** (a) The matched construction drops $|\text{corr}(\text{SUPPORT}, \text{OVERLAP})|$ from 0.99 (extractive NQ-Swap) to 0.50 (model-free cells A/C/D), and would reach ≈ 0 with the paraphrase cell B (projected; cell B has zero yield in this release). (b) Mean NLI entailment by *clean* cell: on the wrong-question trap (C), where the answer is present but unsupported, NLI is still pulled toward “supported.”

Confidence purge. We regress each hidden dimension on confidence covariates (max-softmax, answer log-prob, activation norm) and re-probe the residual, reporting the fraction of probe signal absorbed per axis.

Cross-family transfer. Families differ in hidden size, so we fit a semi-orthogonal Procrustes map between two models’ residual spaces on a shared calibration split, transport a source-trained probe, and measure target AUROC. The calibration/evaluation split is a *GroupKFold* on `ITEM_ID` (items in the calibration set are disjoint from the evaluation set), so the map is never fit and tested on near-duplicate rows of the same item; transported-probe polarity is fixed on the target calibration fold. We compare to an unconstrained ridge map, the anchor-projection ACS baseline (Kim and Han, 2026), a within-family upper bound, and a random-direction floor (also scored without the polarity flip, so it sits at chance rather than being inflated above 0.5).

5 Experiments

Setup. Four instruction-tuned families: Llama-3.1-8B, Qwen2.5-7B, Mistral-7B-v0.3, Gemma-2-9B. CtrlPairs: 1,200 NQ-Swap items $\rightarrow$ 3,593 matched statements (cells A/C/D). RAGTruth: 2,000 test responses with response-level hallucination labels (Niu et al., 2024). Baselines: NLI (DeBERTa-v3-base-mnli-fever-anli (He et al., 2023)) in two variants—(i) *whole-response*: a single (context, declarativized-answer) entailment

query (deliberately simple and conservative for multi-sentence responses); and (ii) *sentence-level*, a SummaC-style decomposition (Laban et al., 2022) that entailment-checks each response sentence against every context sentence, takes the max over context, and aggregates over response sentences by mean (SummaC-ZS) or min (for tractability we cap context at 60 sentences and pairs at 256 tokens, which can only lower SummaC and is thus conservative)—plus max-softmax confidence and lexical overlap. Everything is API-free; the NLI baselines use the small DeBERTa model and all activation extraction runs on one A100-40G.

5.1 A grounding axis exists beyond lexical overlap

Leakage-controlled protocol. Cells A/C/D of an item share the original context c_{org} verbatim (A and C are the same context with a different question), so a plain row-level split would place near-duplicate rows of one item on both sides of the train/test boundary. We therefore use *GroupKFold* on `ITEM_ID` throughout: all statements derived from one NQ-Swap item fall in the same fold, for both the C1 cross-validation and the Procrustes calibration/evaluation split (§5.3). We resolve each estimated axis’s sign on the *train* fold only (never on the test fold), so test AUROC can fall below 0.5 and is not floored by a $\max(\text{AUC}, 1 - \text{AUC})$ flip. To avoid selecting the layer on the very number we report, we fix the analysis layer *a priori* at relative depth 0.5 (the

nearest cached layer to $0.5 L_{\text{tot}}$: $L=14/16/16/21$ for Qwen/Mistral/Llama/Gemma), matching the relative-depth convention already used for C3/C4 and for Fig. 2; we additionally report the mean $\pm$ sd across the mid-layer band (0.4–0.6 depth) to show the result is not a single-layer artifact. Confidence intervals are item cluster bootstraps (resampling ITEM_ID with replacement).

Table 1 reports support decodability within the answer-present stratum $O=1$ (cells A vs. C), where lexical overlap is held constant. Two caveats bound this number. First, within $O=1$ the support contrast varies *only the question* (context and answer are held fixed), and our *synthetic* factuality label is *identical* to support by construction—they diverge only in the answer-absent cell D—so $\text{supp}_{O=1}$ and a within-stratum factuality probe decode the same labels and coincide numerically (≈ 0.877). This number isolates grounding from *lexical overlap*, but *not* from factoid-correctness. Second, in this stratum overlap and answer length are themselves uninformative (AUROC 0.500), so surface removal carries no information here; the operative confound is *model confidence*. Answer log-prob alone separates A from C at 0.80–0.86 and NLI at 0.75; after regressing the hidden states on log-prob and max-softmax the support contrast drops to 0.70–0.78 (Table 1), which we take as the primary confound-controlled estimate (0.87 is a raw upper bound). A cross-strata control that fixes the question at training time (train A vs. D , evaluate A vs. C) still transfers at 0.62–0.78—weakest on Qwen and Gemma—confirming a real but model-dependent grounding component. Because 23.2% of cell-C items are donor-collision mislabels (§3), $\text{supp}_{O=1}$ is a *lower* bound: removing them (at the same a-priori layers) lifts it to 0.99 (0.988–0.999). We therefore claim an overlap-independent, confidence-controlled grounding signal at 0.70–0.78, and we do *not* claim a grounding axis geometrically distinct from factuality (the 51–67° support–factuality angle is a contrast among only three cell-means and, on matched binary partitions of those means, is not by itself evidence of distinctness).

5.2 No confidence asymmetry

We hypothesized that confidence absorbs the factuality axis more than the faithfulness axis, which would offer a way to separate them. The measured asymmetry is small and inconsistent in sign across the families and layers we examined (it does not favour the factuality axis), so we report a null

model	L	$\text{supp}_{O=1}$	95% CI	+conf	cross
Qwen2.5	14	0.872	[.859,.885]	0.70	0.62
Mistral	16	0.874	[.860,.887]	0.78	0.78
Llama	16	0.876	[.863,.890]	0.72	0.78
Gemma	21	0.878	[.864,.892]	0.74	0.75

Table 1: **C1 (confound-controlled)**. Layer fixed a priori at relative depth 0.5; GroupKFold on ITEM_ID; polarity resolved on the train fold. $\text{supp}_{O=1}$ is grouped-CV support AUROC within the answer-present stratum (raw upper bound; item cluster-bootstrap 95% CI). *+conf-purge*: after residualizing the hidden states on confidence (log-prob, max-softmax)—our primary estimate. *cross-strata*: train A vs. D (question fixed), evaluate A vs. C . Within $O=1$, support and the synthetic factuality label coincide, so we make no distinctness claim.

result and do not rely on confidence-purging to distinguish the two notions.

5.3 Grounding directions align across families up to a linear map

We ask whether the grounding direction occupies the *same* geometry across families up to a linear map. The direct evidence is geometric: the Procrustes-mapped source direction is nearly collinear with the target family’s *native* direction (transported-to-native cosine ≈ 0.98 ; Table 2), well above anchor projection. The cosine is far above its null: a random or label-shuffled source direction transported through the *same* fitted map is essentially orthogonal to the native direction (cosine 0.01 and 0.10 respectively), so 0.976 is not an artifact of the shared-stimulus geometry. We deliberately do *not* headline the transferred-probe AUROC (0.906): it equals the within-family bound (0.905) and is therefore ceiling-saturated (a transported probe cannot beat the native one, so AUROC has no headroom to register loss); scored with polarity-honest (signed) AUROC, random and shuffled-label directions through the map sit at chance (0.51/0.51). Critically, the factuality axis transports *identically* (cosine 0.99, AUROC 0.915), so what aligns is a shared low-rank contrast defined by the matched stimuli, not a grounding-specific axis. We therefore conclude that grounding directions *align across these four families up to a linear map* (the cosine evidence is robust to the nulls), and explicitly avoid claims of near-lossless or universal transfer.

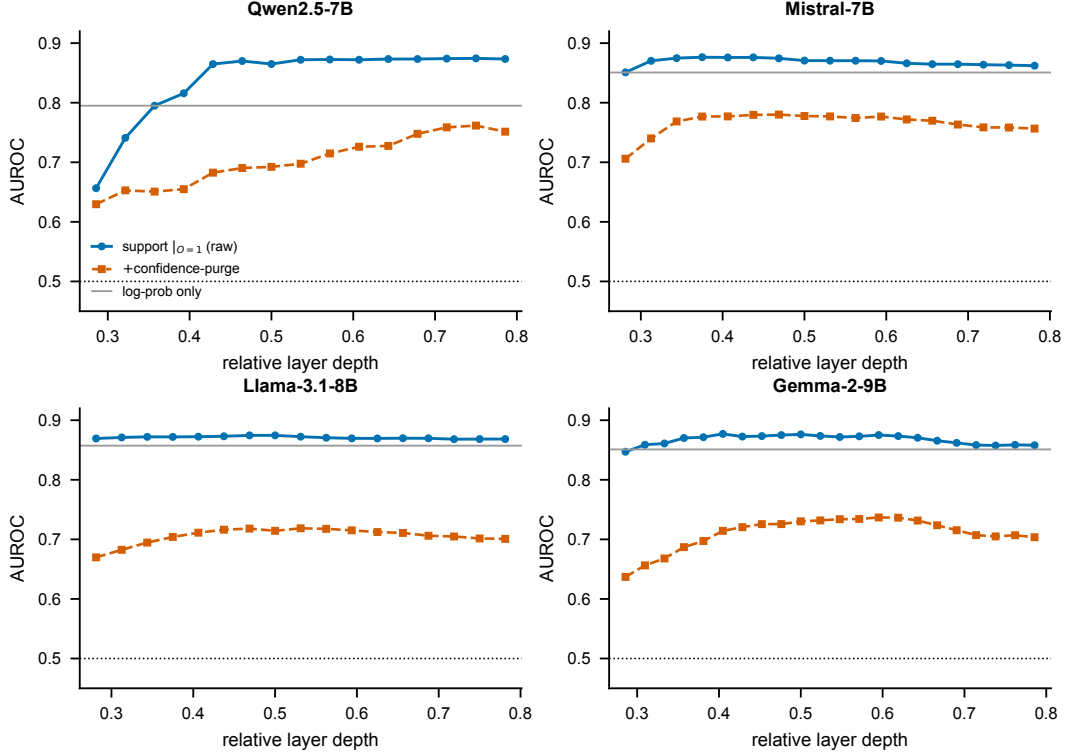


Figure 2: **A grounding axis exists beyond overlap, peaking mid-layer.** Cross-validated AUROC vs. relative layer depth for support within the answer-present stratum ($O=1$, where overlap is uninformative), support after removing surface covariates, and factuality. The overlap-independent support signal is high and stable through mid layers on all four families.

axis	cosine	AUROC	within	rand-map	rand-lab
SUPPORT	0.976	0.906	0.905	0.519	0.506
FACTUALITY	0.990	0.915	0.911	—	—

Table 2: **C3 (cosine-based, leakage-controlled).** Mean over the 12 cross-family ordered pairs, GroupKFold on ITEM_ID (3 seeds), all columns single-protocol. *cosine*: transported-to-native cosine (the alignment evidence; non-saturating). A random / label-shuffled source direction transported through the *same* map has cosine only 0.01 / 0.10 with the native direction, so 0.976 is far above its cosine null. *AUROC/within*: transferred-probe vs. within-family AUROC—equal, hence ceiling-saturated. *rand-map/rand-lab*: random / shuffled-label source directions through the map, scored with signed (polarity-honest) AUROC, sit at chance (0.51). Factuality aligns identically, indicating a shared contrast rather than a grounding-specific axis.

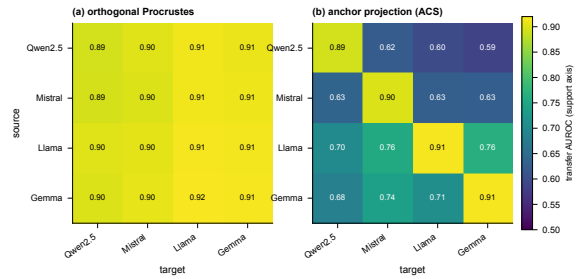


Figure 3: **Grounding directions align across families up to a linear map.** Source→target transfer AUROC of the support direction. (a) Orthogonal Procrustes is uniformly high on- and off-diagonal; we read this as alignment (cosine 0.98, Table 2), not as lossless transfer, since the AUROC is ceiling-saturated. (b) Anchor projection (ACS) matches it only within-family and degrades sharply across families.

5.4 Synthetic axes do not transfer to real hallucinations

We learn d_S on CtrlPairs and apply it to RAGTruth responses of the same model, predicting the human faithfulness label (Table 3). Here we keep the *support-polarity* sign fixed from CtrlPairs and

score without the test-fold flip, because flipping per-test inflates a near-chance axis; CtrlPairs and RAGTruth share no items, so there is no leakage in this transfer. Scored honestly, the label-free synthetic axis reaches only 0.54–0.66 AUROC (and drops toward chance for the weakest family)—below NLI (0.673), model confidence (0.62–0.72),

model	label-free			NLI	overlap
	synth- d_S	conf	in-dom [†]		
Qwen2.5	0.657	0.723	0.766	0.673	0.753
Llama	0.591	0.623	0.816	0.673	0.753
Gemma	0.541	0.686	0.800	0.673	0.753

Table 3: **C4 (leakage-controlled)**. RAGTruth hallucination detection (AUROC). The label-free synthetic grounding axis (synth- d_S , support polarity fixed from CtrlPairs, a-priori mid layer, scored without the test-fold flip) is below NLI, confidence, and lexical overlap, and for the weakest family (Gemma, 0.541) is barely above chance. [†]in-domain probe uses RAGTruth human labels (not label-free).

and plain lexical overlap (0.753), which is the single best predictor on RAGTruth. Removing the polarity flip can only lower these numbers, so the C4 negative is if anything understated by the original table. Thus on real data overlap still dominates, and even NLI does not escape it. A more sophisticated sentence-level NLI baseline (SummaC-style, max entailment over context sentences) does no better: pooled 0.614/0.664 and within-task macro 0.596/0.579 (mean/min), essentially tied with the whole-response NLI (0.582, within sampling noise) and still well below overlap (Table 4)—consistent with RAGTruth hallucinations often being abstractive or reasoning-level errors that survive sentence-by-sentence entailment, on which a small off-the-shelf NLI model has limited headroom. An *in-domain* probe (5-fold grouped-CV L_2 -logistic on the same mid-layer states, target=human faithfulness label) does recover a usable signal: AUROC 0.77/0.82/0.80 for Qwen/Llama/Gemma (item-bootstrap 95% CIs in Table 3). It exceeds our NLI baseline (0.673) on all three families—though that baseline scores each multi-sentence response with a single whole-response entailment check and is conservative (§5); against the strong overlap baseline (0.753) it wins on Llama and Gemma (paired bootstrap $p<0.001$) but only ties on Qwen (+0.01, $p=0.17$). Residualizing lexical overlap out of the features first drops it to 0.60–0.66, so much of even this signal is overlap. A real grounding signal therefore exists in the internals, but it is a *different direction* from the synthetic axis and is recoverable only with in-domain human annotation.

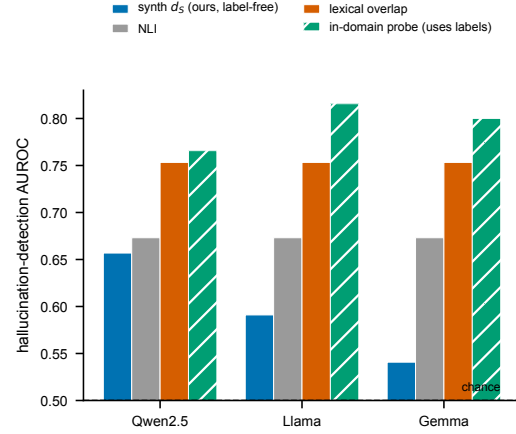


Figure 4: **The synthetic axis does not transfer to real hallucinations.** On RAGTruth, the label-free synthetic support axis (ours) sits below NLI and below plain lexical overlap, the strongest baseline; only an in-domain probe trained on human labels beats NLI. Axis starts at chance (0.5).

signal	Data2txt	QA	Summary	macro
NLI, whole-resp.	0.547	0.567	0.633	0.582
NLI, sent. (mean)	0.611	0.557	0.620	0.596
NLI, sent. (min)	0.521	0.591	0.626	0.579
overlap	0.605	0.686	0.649	0.647
faithful-rate	0.357	0.823	0.795	—

Table 4: **C4, per task type.** Within-task RAGTruth AUROC. The pooled numbers in Table 3 overstate every signal because faithful base rates differ sharply across tasks (bottom row): overlap’s pooled 0.753 falls to macro 0.647, NLI’s 0.673 to 0.582. A more elaborate sentence-level SummaC-style NLI baseline (max over context sentences, then mean or min over response sentences) does no better (macro 0.596/0.579). Within a task, every label-free signal is weak (≈ 0.6); overlap is merely the least-bad.

6 Discussion and Limitations

Our results separate two questions usually merged. *Does* a signal that is not lexical overlap exist in the internals? Yes—within the overlap-controlled stratum, support is decodable at 0.70–0.78 after purging model confidence, and the underlying direction aligns across four model families up to a linear map. *Is* that direction a deployable, label-free hallucination detector? No—it does not transfer to RAGTruth, where lexical overlap remains the strongest single-pass baseline (and even that edge is largely a between-task base-rate artifact: Table 4). The gap between a clean controlled signal and real-world utility is the main lesson, and it cautions

against reading deployment claims from synthetic-stimulus probes. We are explicit about what we do *not* show: (i) within the answer-present stratum support and our synthetic factuality label coincide, so we cannot claim a grounding axis geometrically distinct from factoid-correctness, and even the confidence-purged 0.70–0.78 removes only the *linear* confidence component, so it remains an upper bound on isolated grounding; (ii) cell B has zero yield, so the fully-orthogonalized ($\text{corr} \approx 0$) design is not instantiated and all analyses use cells A/C/D at $\text{corr} = 0.50$ with overlap stratification; (iii) the cosine alignment is measured on shared stimuli, so it shows alignment of a shared low-rank contrast, not a fully unpaired transfer; (iv) RAGTruth covers three task types and one annotation scheme, the negative is on 3/4 families (Mistral’s RAGTruth extraction OOM’d), and stronger internals detectors (e.g. ReDeEP (Sun et al., 2025)) and sampling-based methods are out of our single-A100 budget—so we show that *this* synthetic axis fails, not that internals cannot work. Whether any label-free signal can decisively beat lexical overlap on real hallucinations is the natural next question.

Evaluation rigor. Because cells A/C/D of an item reuse the same context, all cross-validation and all Procrustes calibration/evaluation splits group on ITEM_ID (GroupKFold), so no near-duplicate rows cross the train/test boundary; estimates change negligibly from the ungrouped numbers (e.g. C1 $\text{supp}_{O=1}$ shifts by ≤ 0.01 , C3 raw transfer stays ≈ 0.91), so the positive findings are not leakage artifacts. We resolve axis polarity on the train (or calibration) fold only; the test-fold $\max(\text{AUC}, 1 - \text{AUC})$ convention floors a non-predictive direction above 0.5 (it inflates a true-chance random direction to ≈ 0.54 –0.59), which matters for the *weak* numbers—the C3 map-aware nulls and the synthetic axis on RAGTruth—so we report those without the flip. The C1 analysis layer is fixed a priori at relative depth 0.5 (as C3/C4 already do) rather than argmax -selected on the reported metric. All confound-controlled estimates (confidence-purge, cross-strata, in-domain) and the C3 cosine and nulls are released as a single cached-feature script.

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